Information Technology & Business Analysis

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Introduction to Information Technology

- Information Technology (IT) refers to the use of computers, software, networks, and data systems to collect, store, process, and disseminate information.

Definition of Information Technology

- Concerned with acquisition, processing, storage, and distribution of information
- Supports organizational and societal needs

IT vs Computer Science

- IT: Focus on implementation, management, and support of systems
- Computer Science: Focus on algorithms, computation, and software design

Importance of Information Technology

- Automation of operations
- Efficient communication
- Improved decision-making
- Global connectivity

Applications of IT

- Healthcare
- Banking & Finance
- Education
- Business & Entertainment

Core Components of IT

- Hardware
- Software
- Networking
- Data
- People and Processes

Computer Hardware – Overview

Computer hardware refers to the **physical and tangible parts** of a computer system.

- **Examples:**

Desktop computers

Networking devices

- **Functions of Hardware:**

Accepts input

Processes data

Produces output

Stores information

Input Devices

- Input devices allow users to **enter data and instructions** into a computer system.
- **Examples:**

Keyboard

Mouse

Scanner

Webcam

Output Devices

- Output devices present **processed information** to the user in a usable form.
- **Examples:**

Monitor

Printer

Speakers

Projector

Central Processing Unit (CPU)

- The **Central Processing Unit (CPU)** is known as the *brain of the computer* because it controls and executes all instructions.
- **Main Components of CPU:**
- **Control Unit (CU):**
  - Directs all operations of the computer
  - Controls the flow of data between memory and processor
- **Arithmetic Logic Unit (ALU):**
  - Performs arithmetic operations (addition, subtraction, etc.)
  - Performs logical operations (AND, OR, comparisons)
- **Registers:**
  - Small, high-speed memory inside the CPU

Computer Memory

- Memory stores data, instructions, and results required for processing.
- **Primary Memory:** RAM(Random Access Memory), ROM(Read Only Memory)
- **Secondary Memory:** HDD, SSD

Storage Devices

- **Types of Storage Devices:**

Magnetic Storage: Hard Disk Drives (HDD)

Solid State Storage: SSD, Pen drives

Optical Storage: CD, DVD, Blu-ray

- **Key Advantages:**

Large storage capacity

Data backup and recovery

Computer Software – Overview

- Software is a **set of instructions or programs** that tells the hardware what to do.
- **Functions of Software:**

Controls hardware operations

Enables user interaction

Performs specific tasks

System Software

- System software manages computer hardware and provides a platform for application software.
- **Examples:**

Operating Systems: Windows, Linux, macOS

Device Drivers

Language Translators (Compiler, Interpreter)

- **Functions:**

Memory management

Process management

File system management

Application Software

- Application software helps users **perform specific tasks** such as document creation, calculation, and communication.
  It is designed for **end users** and runs on top of system software.
- **Examples:** Word processors, spreadsheets, accounting software
  Purpose: Improve productivity and simplify user tasks

Utility Software

- Utility software is used to **maintain, protect, and optimize** computer systems.
  It ensures smooth system performance and data safety.
- **Examples:** Antivirus, backup tools, disk cleanup utilities
- **Purpose:** System security, performance improvement, data recovery

Computer Networking

- Computer networking refers to the **interconnection of computers** to share data, hardware, and services.
  It enables communication and collaboration between users.
- **Uses:** File sharing, printer sharing, email
- **Benefit:** Cost reduction and efficient resource utilization

Types of Networks

- **LAN:** Small area like office or campus
- **MAN:** Covers a city
- **WAN:** Covers large areas (countries/continents)
- **PAN:** Personal devices (Bluetooth, mobile)

Internet

- The Internet is a **global network of interconnected networks** that enables information sharing worldwide.
  It supports communication, business, and digital services.
- **Services:** Email, web browsing, cloud services
  Importance: Backbone of the digital economy

Network Topology

- **Network topology** is the **layout of devices in a network** and defines how systems are connected and how data flows.
- **Types:**

Bus: Single cable; low cost, less reliable

Star: Central hub/switch; easy to manage

Ring: Circular connection; sequential data flow

Mesh: Multiple connections; high reliability

Networking Devices

Networking devices are used to **connect computers and control data flow** in a network.

- **Router:** Connects different networks and directs data packets
- **Switch:** Connects devices within a LAN efficiently
- **Hub:** Broadcasts data to all connected devices
- **Modem:** Converts digital signals to analog and vice versa

Database Management System (DBMS)

- A **DBMS** is software that **stores, organizes, and manages structured data** efficiently.
- Reduces data redundancy
- Ensures data consistency and security
- Allows multiple users to access data
- **Examples:** MySQL, Oracle, SQL Server

Importance of DBMS

DBMS is important because it provides:

- **Data accuracy and consistency**
- **Data security** through access control
- **Efficient data retrieval**
- **Scalability** for large databases

Information Security

Information security focuses on **protecting data and IT systems** from unauthorized access, misuse, or damage.

- Ensures confidentiality, integrity, and availability
- Protects organizational and personal data
- Essential for trust and compliance

Cyber Threats

Cyber threats are **malicious activities** that aim to damage, steal, or disrupt data and systems.

- **Virus:** Attaches to files and spreads
- **Malware:** Harmful software
- **Phishing:** Fake messages to steal data
- **Ransomware:** Locks data for ransom

Security Measures

Security measures are used to **protect systems and data** from cyber threats.

- **Firewalls:** Control network access
- **Encryption:** Protects data during transmission
- **Authentication:** Verifies user identity

Cloud Computing

Cloud computing provides **on-demand computing resources** over the Internet.

- Eliminates need for local infrastructure
- Accessible from anywhere
- Used by businesses for storage and applications

Cloud Service Models

Cloud services are offered in three main models:

- **IaaS:** Infrastructure like servers and storage
- **PaaS:** Platform for application development
- **SaaS:** Ready-to-use software applications

Benefits of Cloud Computing

Cloud computing offers several advantages:

- **Scalability:** Resources as needed
- **Cost efficiency:** Pay-as-you-use
- **Accessibility:** Anywhere access
- **Reliability:** Data backup and recovery

Emerging IT Technologies

Emerging technologies are **new and advanced technologies** that are transforming how organizations operate.

- **Artificial Intelligence (AI):** Automates tasks and decision-making
- **Internet of Things (IoT):** Connects smart devices via the Internet
- **Blockchain:** Secure and decentralized data storage
- **Big Data:** Analysis of very large datasets

Role of IT in Modern Business

Information Technology supports **business operations and management**.

- Enables **automation of processes**
- Improves **efficiency and productivity**
- Supports systems like **ERP and CRM**
- Helps in **data analysis and reporting**

Enterprise Resource Planning (ERP)

Enterprise Resource Planning (ERP) is an **integrated software system** used to manage an organization's core business processes through a **centralized database**.

- Integrates different departments into one system
- Enables real-time data sharing
- Improves coordination and efficiency
- **Examples:** SAP, Oracle ERP, Microsoft Dynamics

Core Modules of ERP

ERP systems consist of **modules**, each handling a specific business function.

- **Finance & Accounting:** Billing, budgeting, reporting
- **Human Resource Management (HRM):** Payroll, recruitment
- **Inventory Management:** Stock control
- **Sales & Marketing:** Orders, billing
- **Supply Chain Management (SCM):** Procurement and logistics

Benefits of ERP

ERP provides several organizational advantages:

- Automation of business processes
- Accurate and real-time reporting
- Better decision-making
- Reduced operational costs
- Improved inter-department coordination

ERP vs CRM

Aspect	ERP	CRM
Purpose	Manage internal processes	Manage customer relationships
Focus	Operations & efficiency	Sales & customer satisfaction
Users	Finance, HR, Operations	Sales & Marketing teams

Business Strategy and IT Strategy

- **Business Strategy:** Defines organizational goals and direction
- **IT Strategy:** Defines how technology supports business goals

Alignment Capability

Alignment capability refers to an organization's ability to **synchronize business strategy and IT strategy** so they support each other effectively.

- Requires strong communication between business and IT teams
- Involves leadership, governance, and flexibility
- Ensures technology investments deliver business value

Business–IT Alignment Cycle

This cycle explains how **business and IT continuously support each other**.

- Business strategy defines goals
- IT strategy is designed to support these goals
- Alignment ensures both remain coordinated
- Improved performance influences future strategies

Types of Data

- Structured
- Semi-structured
- Unstructured

Structured Data

Structured data is data that is **highly organized** and stored in a **fixed format** such as rows and columns.

- Uses predefined schema
- Stored in relational databases
- Easy to query using SQL
- **Examples:** Excel sheets, bank records, employee databases

Semi-Structured Data

Semi-structured data does not follow a strict table format but has **some level of organization**.

- Uses tags or key–value pairs
- Flexible structure
- Common in web-based systems
- **Examples:** JSON files, XML files, emails

Unstructured Data

Unstructured data is data that has **no predefined format or schema**.

- Not stored in rows and columns
- Difficult to process using traditional databases
- Requires advanced techniques like NLP and image processing
- **Examples:**
  Text documents, images, audio files, videos, social media posts

Data Sources

Data sources are the **origins from which data is obtained** for analysis and decision-making.

- **Types of Data Sources:**

Primary Data Sources:
Data collected directly from original sources.

Secondary Data Sources:
Data collected from existing records and publications.

Data Collection

- **Data collection** is the process of **gathering data from various sources** for analysis and decision-making.
- **Qualitative Data:** Opinions, feedback, descriptions
- **Quantitative Data:** Numerical data such as sales and income
- **Sources:**
  Primary data and secondary data

Primary Data Collection

Primary data is **first-hand data** collected directly for a specific purpose.

- **Methods:** Surveys, interviews, observations, experiments
- Highly accurate and relevant
- Time-consuming and costly

Secondary Data Collection

Secondary data is data collected from **existing sources**.

- **Sources:** Books, journals, reports, websites, government data
- Saves time and cost
- May be outdated or less specific

Data Life Cycle Management

Data Life Cycle Management (DLM) manages data **from creation to disposal**.

- **Stages of DLM:**
- Data creation
- Storage
- Usage
- Maintenance
- Archival or deletion
- **Importance:**
  Ensures data quality, security, and regulatory compliance.

Business Analyst – Introduction

A **Business Analyst (BA)** acts as a **link between business needs and IT solutions**.

- Understands business problems and requirements
- Analyzes data and processes
- Recommends suitable solutions

Responsibilities of Business Analyst

A Business Analyst is responsible for:

- Identifying business needs and problems
- Gathering and documenting requirements
- Analyzing data and business processes
- Communicating with stakeholders
- Supporting solution implementation

Skills of a Business Analyst

Key skills required for an effective Business Analyst include:

- Analytical and critical thinking
- Strong communication skills
- Problem-solving ability
- Documentation and modeling skills
- Basic understanding of IT and databases

Tools Used by Business Analysts

- Excel
- SQL
- BI tools
- JIRA
- Visio

Business Analysis Methodologies

- Business analysis uses **structured methodologies** to manage projects and requirements effectively.
- Helps plan, analyze, and control projects
- Ensures clear communication between stakeholders
- Two major methodologies:
  - **Waterfall**
  - **Agile**

Waterfall vs Agile

Basis	Waterfall	Agile
Approach	Sequential	Iterative
Flexibility	Low	High
Customer Involvement	Limited	Continuous
Delivery	End of project	Frequent increments

When to Use Waterfall

Waterfall methodology is suitable when:

- Requirements are **clear and stable**
- Project scope is **well-defined** from the start
- **Strict documentation** and regulatory compliance are required
- Customer involvement is **limited**
- Fixed **budget and timeline** are important

When to Use Agile

Agile methodology is suitable when:

- Requirements are **likely to change**
- Continuous **customer feedback** is required
- Fast and **incremental delivery** is needed
- Project is **complex or uncertain**
- Cross-functional teams work closely together

Stakeholder Management

Stakeholder management is the process of **identifying, analyzing, and engaging stakeholders** involved in a project.

- Stakeholders include customers, employees, managers, and regulators
- Helps manage expectations and reduce risks
- Improves communication and project success

Scenario Analysis

Scenario analysis is a technique used to **evaluate different possible future situations** to support decision-making.

- Considers **best-case, worst-case, and most-likely** scenarios
- Helps identify **risks and opportunities**
- Supports strategic planning and forecasting
- **Example:**
  Analyzing the impact of changes in sales, cost, or demand on profit.

Benefits of Scenario Analysis

Scenario analysis provides several advantages for organizations:

- Helps identify **potential risks and opportunities**
- Supports **better strategic and financial planning**
- Improves **decision-making under uncertainty**
- Enables development of **contingency plans**
- Reduces impact of unexpected changes

Data Governance

Data governance refers to the set of **policies, rules, roles, and processes** used to ensure data is **accurate, secure, and properly used**.

- Defines who can access and manage data
- Ensures data quality and consistency
- Supports compliance with laws and regulations

Benefits of Data Governance

Effective data governance provides several benefits:

- Improved data quality and accuracy
- Better and more confident decision-making
- Enhanced data security and privacy
- Clear roles and responsibilities
- Regulatory compliance

Ethical Use of Data

The ethical use of data involves **using data responsibly and fairly**.

- Obtaining user **consent and transparency**
- Avoiding bias and discrimination
- Collecting only **necessary data**
- Protecting privacy and security

GDPR vs CCPA – Overview

GDPR and **CCPA** are data protection laws that give individuals **control over their personal data**.

- GDPR applies in the **European Union**
- CCPA applies in **California, USA**
- Both focus on data privacy and transparency

Conclusion

- Information Technology supports modern business and society
- ERP, data management, and business analysis improve efficiency
- Ethical and governed use of data is essential
- Alignment of business and IT ensures long-term success

Thank You!